

Physics-Integrated Generative Edge-AI Framework for Explainable Aero-Engine Prognostics

Milestone 1 Submission | IEEE IES Generative AI Challenge 2026

1. Abstract

Prognostics and Health Management of aircraft turbofan engines is a critical challenge in aerospace engineering, where accurate Remaining Useful Life (RUL) prediction enables proactive maintenance, reduces unplanned downtime, and improves operational safety. Existing approaches, from recurrent networks to attention-based transformers, predominantly deliver single-point RUL estimates with no uncertainty quantification and no mechanism to verify whether predictions respect the physical laws governing engine degradation. This work proposes a Physics-Integrated Generative Edge-AI Framework trained on the NASA C-MAPSS FD001 benchmark that replaces scalar point prediction with generative counterfactual prognostics. An attention-based encoder produces latent degradation embeddings that condition a physics-constrained conditional VAE to sample multiple plausible future degradation trajectories. Physics-derived constraints including degradation monotonicity and thermodynamic consistency are embedded directly into the training loss rather than applied post-hoc. A dual-head output delivers both a probabilistic RUL distribution and a physics-violation risk score per trajectory. A post-prediction Reliability Index feeds a three-tier gating layer producing ACCEPT, WARN, or REJECT decisions. Explainability is quantified through the novel Counterfactual Physical Consistency (CPC) metric, measuring whether generated trajectories remain physically coherent under shifted operating conditions. The system is deployed via a Streamlit dashboard, FastAPI backend, and an MQTT-connected live digital twin.

2. Literature Review

The following papers are reviewed with emphasis on existing implementations and key research gaps addressed by this work.

- **Saxena et al. (2008), PHM08:** Established the C-MAPSS benchmark through thermodynamic simulation of turbofan degradation. Introduced the asymmetric NASA scoring function. Gap: the thermodynamic physics embedded in the simulator are not re-used as training constraints in downstream models, and no trajectory-level ground truth exists for future degradation paths.
- **Heimes (2008), Zheng et al. (2017):** Applied LSTMs and RNNs to C-MAPSS sensor windows, establishing recurrent baselines. Gap: all outputs are deterministic scalar values with no uncertainty representation, no stochastic trajectory modelling, and no internal mechanism to detect physically implausible predictions.
- **Zhang et al. (2022), IEEE TIM:** Proposed dual-aspect self-attention (temporal and sensor-wise) for RUL prediction, advancing the transformer-based state of the art on C-MAPSS. Gap: output remains a single scalar, explainability relies on post-hoc SHAP attribution, and physics are entirely absent from the model architecture.
- **Dempster et al. (2019), ROCKET:** Demonstrated that random convolutional kernels achieve state-of-the-art time series classification accuracy at a fraction of the cost of deep learning methods. Gap: designed for classification only, with no uncertainty estimates, no generative capability, and no physical priors.
- **Christ et al. (2016), tsfresh:** Proposed scalable parallel feature extraction for industrial time series using non-parametric hypothesis testing. Gap: a preprocessing methodology only, producing engineered features with no physics grounding and no ability to reason over future degradation trajectories.
- **Bayesian RUL approaches (Zhu et al. 2019, Biggio et al. 2021):** Introduced uncertainty quantification via Monte Carlo dropout and deep ensembles, producing credible intervals alongside point estimates. Gap: uncertainty is purely statistical; no physically coherent trajectory generation, no counterfactual reasoning, and no physics-violation detection.
- **Li et al. (2024), physics-informed RUL review:** Identified physics-informed frameworks as an emerging direction, but found that most works apply physics post-hoc or as input features rather than embedding them into the learning objective. Gap: no existing work embeds physics constraints into a generative model's training loss to ensure sampled trajectories are physically coherent by construction.

The literature converges on three unresolved problems this project addresses simultaneously: (1) all mainstream models produce point estimates with no trajectory-level uncertainty; (2) physics knowledge is applied post-hoc rather than embedded in the generative process; and (3) no existing work supports physically grounded counterfactual reasoning over RUL under shifted operating conditions.

3. Proposed Methodology

The pipeline below covers the end-to-end flow from raw sensor ingestion to gated RUL output. Stages 1 and 2 are shared across all modes. Stage 3 introduces a physics-constrained generative model that samples multiple degradation trajectories before converging at the dual-head output. Gating (stage 6) is strictly post-prediction — raw model output is always preserved for analysis.

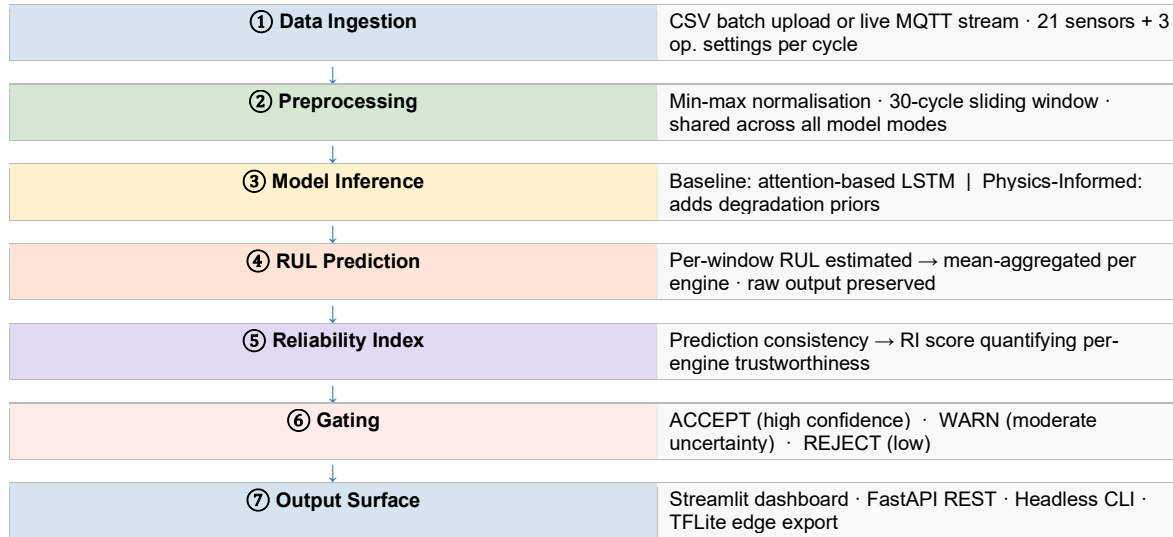


Figure 1. End-to-end generative RUL prediction pipeline with physics constraints and post-prediction reliability gating

4. Dataset Plan

Primary dataset: NASA C-MAPSS FD001

Single operating condition, single fault mode (HPC degradation), 100 train / 100 test engine trajectories. Each run-to-failure trace provides 21 sensor channels and 3 operational settings per time cycle.

Subset	Train	Test	Sensors	Op. Cond.	Fault	Project Use
FD001 (primary)	100	100	21	1	HPC	Training & evaluation
FD002	260	259	21	6	HPC	Multi-condition test
FD003	100	100	21	1	HPC+Fan	Multi-fault eval.
FD004	248	249	21	6	HPC+Fan	Full complexity

Table 1. NASA C-MAPSS subsets and project usage allocation

Preprocessing protocol

- RUL labeling: piecewise linear with 125-cycle cap reflecting the stable early-life operating phase
- Normalization: min-max per feature computed on training set; same parameters applied to test set
- Windowing: 30-cycle sliding window; early-cycle sequences zero-padded at the front
- Sensor selection: near-zero variance sensors (s_1, s_5, s_6, s_10, s_16, s_18, s_19) excluded

Evaluation metrics & validation

- RMSE — symmetric deviation penalty on the FD001 test set
- NASA score function — asymmetric, penalizing early predictions over late (reflects maintenance risk)
- RI acceptance rate — proportion of test engines receiving ACCEPT gate under calibrated thresholds
- Regression baselines stored at first validation run; checked by tests/test_inference_regression.py
- Optional LightGBM backend benchmarked via scripts/benchmark_model_backends.py before any promotion